Mathemagics

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A Field Guide to Random Matrices and Spectral Clustering

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under the supervision of

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Mathemagics: A Field Guide to Random Matrices and Spectral Clustering

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COLOPHON. This book was typeset in LaTeX using the `tufte-book` class as the foundation, with custom theorem environments built on `tcolorbox` and Python code rendered via `minted`. The visual identity is inspired by the Tufte school of book design and by Mykel Kochenderfer's and Tim Wheeler's series of textbooks for MIT Press. Source files and Colab notebooks live alongside the book at `[repo URL placeholder]`.

First draft: April 2026.

Preface

Placeholder content for this section. The preface is a tour for the reader, not the author. A short version follows; the full text will be written once the chapters are filled in.

Who this book is for

This book is written for the curious reader with a basic feel for probability and algebra, who wants to understand why *random matrices* are everywhere in modern data science, from principal component analysis to deep learning, from finance to wireless communications, and who has been told the words "Tracy–Widom," "spiked covariance," or "spectral clustering" without ever being shown how they fit together.

If you have seen a normal distribution and you know what an eigenvalue is, you have enough background to start. Everything else is built up from scratch.

How to read this book

The book supports two reading speeds.

The cliff-notes track. Read the chapter opener (Where we are box), the Intuition boxes, the figures, and the Chapter recap at the end. Skip the

proofs. You will come away with the conceptual picture in roughly 30% of the time.

The deep-diver track. Read everything: definitions, theorems, proofs, examples, simulations, exercises. The book is structured so that every claim is either proved in-line, sketched with a pointer to a full proof in the chapter appendix, or quoted as a standard result with a citation.

The right margin is reserved for asides, small figures, code snippets, and Colab badges. *Asides never carry plot-load*, if you ignore the entire margin, the proofs still work. Read the margins for context, history, motivation, or the occasional small joke.

Placeholder content for this section. Cliff-notes track listing: explicit list of which sections to read. To be written once chapters are filled in.

What you need to know first

- Some idea of what a random variable is, and what a probability density looks like (we'll re-build all of this from scratch in Part I, but having seen it once before will speed things up).
- Comfort with linear algebra at the level of vectors, matrices, eigenvalues, and a dot product. We re-derive the spectral theorem from scratch in Chapter 6.
- Basic Python literacy, enough to read a NumPy snippet and run a Colab notebook.

What you'll be able to do after

After reading this book, you will:

- Read research papers on random matrix theory and follow the central results without getting stuck on definitions.

- Run and interpret simulations of sample covariance matrices, eigenvalue distributions, and Marchenko–Pastur and Tracy–Widom limits.
- Recognize the spiked covariance model and the BBP phase transition in real data.
- Implement spectral clustering and explain *three* different reasons it works.

A note on style

The book is written in third person, in the style of polished research notes that have been opened up to a wider audience. Technical content is rigorous; prose is friendly. Cartoon asides exist. The title is a small joke; the math is not.

Arzaan Singh

New Orleans, 2026-04-27

Acknowledgments

Placeholder content for this section. Acknowledgments will be filled in after the first content pass. Names already secured: Gustavo Didier (advisor), Joseph Genzer (whose honors thesis is the structural seed of this book), the Tulane Department of Mathematics.

Notation

This is the notation we use throughout the book. The table is sorted roughly by frequency of use.

Placeholder content for this section. Notation guide will be expanded as chapters are filled in. The table below is the seed.

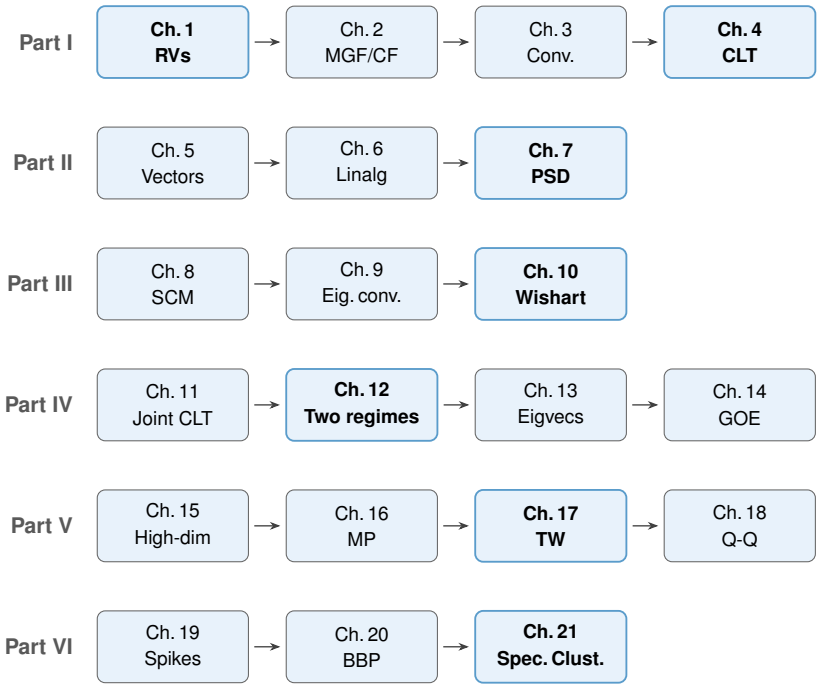
Symbol	Meaning
$\mathbb{R}, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{C}$	real, natural, integer, rational, complex numbers
$\mathbb{E}[X]$	expectation of the random variable X
$\mathrm{Var}(X), \mathrm{Cov}(X, Y)$	variance, covariance
$\mathcal{N}(\mu, \sigma^2)$	normal (Gaussian) distribution with mean μ, variance σ^2
$\mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$	multivariate normal in $\mathbb{R}^p$
$\xrightarrow{\text{a.s.}}, \xrightarrow{P}, \xrightarrow{d}$	convergence almost surely / in probability / in distribution
Σ	population covariance matrix
$\widehat{\Sigma}_n$	sample covariance matrix from n observations
$\lambda_i(M)$	ith eigenvalue of the matrix M (ordered)
$\mathcal{W}_p(n, \Sigma)$	Wishart distribution with n degrees of freedom, scale Σ
W	Gaussian Orthogonal Ensemble (GOE) random matrix limit
$f_{\mathrm{MP}}(\lambda)$	Marchenko–Pastur density
TW_1	Tracy–Widom law of type 1 ($\beta = 1$)
$\langle u, v \rangle$	inner product of vectors u and v
$\lVert M \rVert_F, \lVert M \rVert_{\mathrm{op}}$	Frobenius and operator norms of a matrix M
$\mathrm{diag}(a_1, \ldots, a_p)$	diagonal matrix with the listed entries
I_p	$p \times p$ identity matrix

How the chapters depend on each other

The chart below is a reading roadmap. Each row is one Part; within each row, an arrow "**Ch. A** $\rightarrow$ **Ch. B**" means *read Ch. A first*. Read top-to-bottom to progress through the book.

If you only have time for the spine, follow the cliff-notes track:

Ch. 1 $\rightarrow$ Ch. 4 $\rightarrow$ Ch. 7 $\rightarrow$ Ch. 10 $\rightarrow$ Ch. 12 $\rightarrow$ Ch. 17 $\rightarrow$ Ch. 21.



Bordered nodes are on the cliff-notes spine.

The graph is intentionally lossy, it shows the within-Part reading order, not every cross-reference. When in doubt, the table of contents always works.

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Prelude

An appetizer. Where this book is going, in pictures, with almost no equations.

1

An Appetizer

Where we are

You're starting at the beginning. This chapter is a tour of the destination, random matrix theory and spectral clustering, in pictures, with almost no math. After the tour, the rest of the book builds the machinery to make every picture rigorous.

Placeholder content for this section. The prelude. A short, picture-driven tour of where the book is going. To be written last, after the rest of the book is drafted, so it can promise exactly the punchlines we deliver.

1.1 Random matrices, in two pictures

Placeholder content for this section. Marchenko–Pastur picture; spike-detected-by-PCA picture.

1.2 *Why the journey starts with coin flips*

Placeholder content for this section. Argument that even Marchenko–Pastur eventually reduces to a CLT-flavored statement.

1.3 *What you'll be able to do at the end*

Placeholder content for this section. Concrete checklist matching the preface.

Chapter recap

- *Placeholder content for this section. Three bullet teasers.*

Part I

Foundations of Probability

We start with the bare minimum. What is a random variable, what is its distribution, how do we summarize it, and what happens when we add many of them up? Part I builds the probability vocabulary that the rest of the book leans on.

Chapters in this part: 1. Random Variables 2. Generating Functions 3. Convergence 4. The Central Limit Theorem.

2

Random Variables: A Working Vocabulary

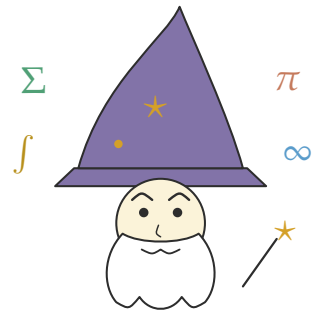
Where we are

This is the first chapter of the book. Everything that follows, moment generating functions, the central limit theorem, sample covariance matrices, eigenvalue fluctuations, and ultimately spectral clustering, is built on top of the language we develop here.

By the end of this chapter the reader will be able to: define random variables and their distributions cleanly, compute expectations and variances of all the running examples (Bernoulli, Binomial, Normal, Exponential), and recognize when two random variables are independent. The chapter assumes only basic high-school algebra and a willingness to draw a few pictures.

Probability is the language● we use when we want to say something precise about a quantity that is uncertain. The outcome of a coin flip, the height of a randomly chosen person, the largest eigenvalue of a sample covariance matrix; all of these vary across repetitions of the experiment, and probability supplies the vocabulary for describing how. The unit of vocabulary is the *random variable*: a single number whose value is unknown until the experiment is run. This chapter builds that vocabulary from scratch and trains the running examples that the rest of the book will lean on.

● MATHEMAGICS



Probability is rigorous magic: we name the unknown, weigh it, and pull conclusions out of it.

2.1 *Sample spaces, outcomes, and the function we call X*

Before we can define a random variable, we need to be precise about what it "randomizes over." ● That requires three pieces:

1. an *experiment*, the act whose result is uncertain;
2. the *sample space* Ω, the set of all possible outcomes of the experiment;
3. a *probability function* $\mathbb{P}$ that assigns a number in $[0, 1]$ to events (i.e., subsets of Ω), with $\mathbb{P}(\Omega) = 1$.

The simplest example is a single coin flip. ● The experiment is the flip; the sample space is $\Omega = \{H, T\}$; and the probability function (for a fair coin) is $\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}$. The outcomes here are just the labels H and T. They are not numbers, they are labels for possibilities.

Often we care less about the labels themselves than about a number we can compute from them. For two coin flips, we may care about the *number of heads*; for the roll of two dice, we may care about the *sum*; for a clinical trial, we may care about whether a patient recovered or not. In each case there is a function that takes an outcome and returns a number. That function is the random variable. ●

Definition 1: Random variable

● Let Ω be a sample space. A *random variable* on Ω is a function $X : \Omega \to \mathbb{R}$ that assigns to each outcome $\omega \in \Omega$ a real number $X(\omega)$.[1]

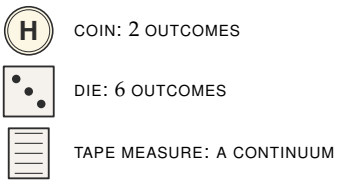
Intuition

Think of X as a *measurement device*. The randomness lives in Ω; it is the underlying experiment whose outcome we cannot predict. The random variable X does not introduce any new randomness; it just tells you what number you record once an outcome has been drawn.

To see what the definition means in practice, take the simplest non-trivial example.

HISTORICAL ASIDE. ● The modern definition of a random variable as a function on a sample space is due to Andrey Kolmogorov, in his 1933 monograph *Foundations of the Theory of Probability*. Before Kolmogorov, "random" was a word in search of a definition.

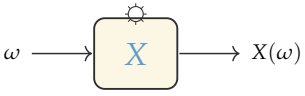
● THREE RUNNING EXPERIMENTS



[1] In a fully rigorous treatment we would also require X to be *measurable*; this technical condition becomes relevant only when Ω is uncountable. Appendix D sketches the measure-theoretic framework for the curious reader.

WHY THIS MATTERS. ● Defining X as a function on Ω, rather than just as "a random number," lets us recover *any* numerical question about the experiment by the same mechanism. The number of heads, the number of tails, whether the first flip matched the second; each is a different function on the same Ω.

● A RANDOM VARIABLE AS A MACHINE



Outcomes go in; numbers come out. The randomness lives in the input, not in the machine.

Example 1: Number of heads in two coin flips

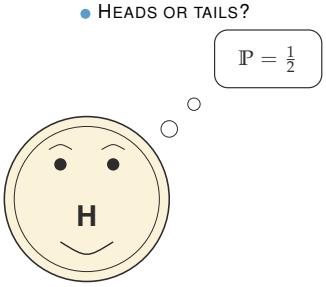
- Flip a fair coin twice in a row. The sample space contains four outcomes,

$$\Omega = \{HH, HT, TH, TT\},$$

each with probability $\frac{1}{4}$. Define the random variable

$$X(\omega) = \text{the number of heads in } \omega.$$

Then $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$, and $X(TT) = 0$. So X takes its values in $\{0, 1, 2\} \subset \mathbb{R}$. Figure 2.1 draws the function: outcomes on the left, sorted by their image; the real number line on the right; arrows from each outcome to its target, with HT and TH visibly converging at 1.



One coin flip is the simplest non-trivial experiment in all of probability.

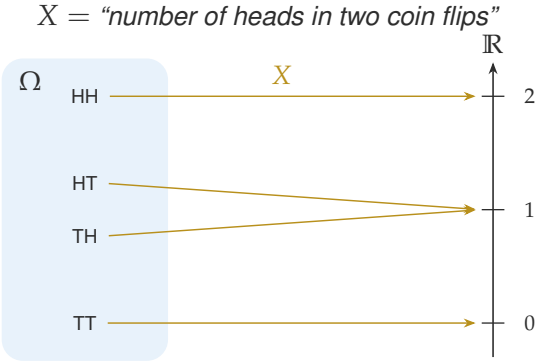


Figure 2.1: The random variable X assigns to every outcome $\omega \in \Omega$ a real number $X(\omega)$. Outcomes that share a value share a target on the line: here HT and TH both map to 1, which is why $\mathbb{P}(X{=}1) = \frac{1}{2}$. Layout adapted from [NRI14].

Before moving to a richer example, two notational conventions are worth pinning down. They will be used everywhere in this book.

Definition 2: Notation: X versus x

Capital letters ($X, Y, Z, \ldots$) denote random variables, the *functions* on Ω. Lowercase letters ($x, y, z, \ldots$) denote particular real numbers, specific values that those random variables *could* take.

So when we write $\mathbb{P}(X = x)$, we are asking: what is the probability that the

random variable X takes the value x? In the two-coin example above,

$$\mathbb{P}(X = 0) = \tfrac{1}{4}, \qquad \mathbb{P}(X = 1) = \tfrac{1}{2}, \qquad \mathbb{P}(X = 2) = \tfrac{1}{4}.$$

The probability $\mathbb{P}(X = 1) = \frac{1}{2}$ comes from the fact that two of the four equally-likely outcomes (HT and TH) map to 1.

A second example, with a richer sample space, exercises the same pattern.

Example 2: Sum of two dice

Roll two fair six-sided dice. ● The sample space contains 36 ordered pairs:

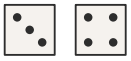
$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\},$$

each with probability $\frac{1}{36}$. Let $X(i, j) = i + j$ be the sum. Then X takes values in $\{2, 3, \ldots, 12\}$, and the probability of each value is the number of (i, j) pairs that sum to it, divided by 36:

$$\mathbb{P}(X = 2) = \tfrac{1}{36}, \quad \mathbb{P}(X = 7) = \tfrac{6}{36}, \quad \mathbb{P}(X = 12) = \tfrac{1}{36}.$$

The full distribution of X ● has the familiar triangular shape: rare on the extremes (only one pair sums to 2), most likely in the middle (six different pairs sum to 7). Figure 2.2 plots it.

● ONE OUTCOME, SUM = 7



One of six pairs that sum to seven; the others are (⚀, ⚅), (⚁, ⚄), (⚂, ⚃), (⚃, ⚂), (⚄, ⚁), (⚅, ⚀).

This is the working pattern. We start from a sample space whose outcomes carry no numerical structure; we define a function X that turns each outcome into a number; and we pull the probabilities forward through X to talk about $\mathbb{P}(X = x)$ or, more generally, $\mathbb{P}(X \in A)$ for any subset A of $\mathbb{R}$.

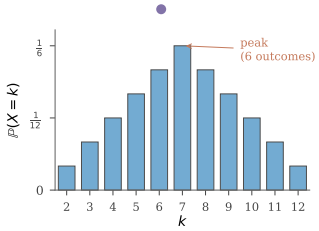


Figure 2.2: The distribution of X in **??** 2. The triangular peak at $k = 7$ is no coincidence: of the 36 equally-likely pairs (i, j), exactly 6 sum to 7.

Brief warm-ups

Hint. ● How many outcomes do you get from flipping three coins? Try listing them in order, then count those with exactly k heads.

Warm-up 2.1 ●. *Three coins are flipped, all fair, all independent.* ● *Write down the sample space Ω. Define the random variable X to be the number of heads. List $X(\omega)$ for each $\omega \in \Omega$, and compute $\mathbb{P}(X = k)$ for $k = 0, 1, 2, 3$.*

► **Reveal answer**

Hint. ● A standard deck has 52 cards, half of them red.

Warm-up 2.2 ●. *A single card is drawn from a standard 52-card deck.* ● *Define a random variable Y that equals 1 if the card is red and 0 if the card is black. What is the sample space, and what is $\mathbb{P}(Y = 1)$?*

► **Reveal answer**

Hint. ● Out of the 36 ordered pairs in Ω, how many have $i = j$?

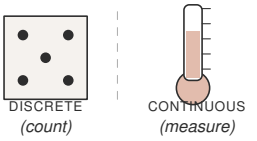
Warm-up 2.3 ●●. *Re-do* **??** *2 but with a different random variable:* ● *let $Z(i, j)$ be 1 if $i = j$ and 0 otherwise, so Z records whether the two dice show the same face. Compute $\mathbb{P}(Z = 1)$.*

► **Reveal answer**

2.2 *Distributions: PMFs, PDFs, and CDFs*

We have a random variable ●; how do we summarize its behavior compactly? The full set of probabilities $\mathbb{P}(X \in A)$ for every $A \subseteq \mathbb{R}$ is too big to write down directly. We need a single object, a function on $\mathbb{R}$, that contains the same information. There are two natural answers depending on what kind of values X takes, plus a third object that handles both at once.

● DISCRETE VS CONTINUOUS



Two flavours of randomness, two parallel ways to summarize.

A random variable X is *discrete* if it takes values in a finite or countable set: integers, indicator labels $\{0, 1\}$, the natural numbers. The number of heads in two coin flips and the sum of two dice are both discrete. A random variable X is *continuous* if its values lie in an interval (or all of $\mathbb{R}$): a length, a temperature, a time. Discrete and continuous random variables share the same definition but require different summaries because of how their probability is allocated. We work through the discrete case first, then the continuous case, then a single unifying object that handles both.

Discrete: the probability mass function

The simplest summary of a discrete random variable lists the probability of every value it can take. That list is the PMF.

Definition 3: Probability mass function (PMF)

Let X be a discrete random variable. The *probability mass function* of X is the function $p_X : \mathbb{R} \to [0, 1]$ defined by

$$p_X(x) \;=\; \mathbb{P}(X = x).$$

A PMF satisfies $p_X(x) \geq 0$ for every x, and $\sum_x p_X(x) = 1$, where the sum is over all values x that X can take.

Intuition

The PMF places a *weight* at each value the random variable can take, and the weights sum to 1. Visualised as a bar chart, the height of the bar at x

is the probability of seeing exactly that value.

Four named discrete distributions show up repeatedly in the rest of this book. Each one below comes with a short story (what it models), its PMF, and a numerical example. The mean and variance are stated as facts; the techniques for computing them are introduced in Section 2.3.

The simplest discrete distribution has just two outcomes.

Bernoulli in the wild

Anywhere a single trial has just two outcomes: a coin lands heads or tails; a clinical patient recovers or does not; an ad impression results in a click or it does not; an exam answer is right or wrong; a packet of internet data arrives or is lost.

Example 3: Bernoulli(p)

A *Bernoulli random variable* $X \sim \text{Bernoulli}(p)$ takes the value 1 with probability p and the value 0 with probability $1-p$. Its PMF is

$$p_X(1) = p, \qquad p_X(0) = 1 - p.$$

Mean: p. Variance: $p(1-p)$, largest at $p = \frac{1}{2}$ (the maximally uncertain coin).

The card-colour random variable Y from Section 2.1 is a Bernoulli with $p = \frac{1}{2}$. So is the indicator of any single yes/no event.

If we run n independent Bernoulli(p) trials and count the successes, we get the most useful discrete distribution in statistics.

Binomial in the wild

The number of: heads in n coin flips; voters in a sample of n who support a candidate; defective items in a batch of n products; successful free throws out of n attempts; ad clicks out of n impressions.

Example 4: Binomial(n, p)

A *Binomial random variable* $X \sim \text{Bin}(n, p)$ counts the number of successes in n independent Bernoulli(p) trials, so $X = X_1 + \cdots + X_n$ where each $X_i \sim \text{Bernoulli}(p)$. Its PMF is

$$p_X(k) \;=\; \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

The factor $\binom{n}{k}$ counts the distinct ways to arrange k successes among n trials; $p^k(1-p)^{n-k}$ is the probability of any one such arrangement. Mean: np. Variance: $np(1-p)$.

A worked numeric. The probability of getting exactly 3 heads in 5 flips of a

fair coin is

$$\mathbb{P}(X = 3) = \binom{5}{3} \cdot \left(\tfrac{1}{2}\right)^3 \cdot \left(\tfrac{1}{2}\right)^2 = 10 \cdot \tfrac{1}{32} = \tfrac{5}{16}.$$

Where Bernoulli answers *yes/no* and Binomial answers *how many out of n*, the Geometric answers *how long do I have to wait*.

GEOMETRIC IN THE WILD

The number of trials until something happens for the first time: flips until the first head; job applications until the first offer; guesses until cracking a four-digit PIN; batters until the first hit.

Example 5: Geometric(p)

A *Geometric random variable* $X \sim \text{Geometric}(p)$ counts the number of independent Bernoulli(p) trials needed to get the first success. Its PMF is

$$p_X(k) \;=\; (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

The reasoning: the first $k-1$ trials must all fail (probability $(1-p)^{k-1}$) and the k-th must succeed (probability p). Mean: $1/p$. Variance: $(1-p)/p^2$. For a fair die, the expected number of rolls until the first six is 6.

If a rare event happens many times independently over a fixed window, the count is Poisson.

POISSON IN THE WILD

The count of rare events in a fixed window: phone calls per hour at a help desk; mutations per generation in a single gene; cosmic rays per second hitting a detector; typos per page in a manuscript; soccer goals per match.

Example 6: Poisson(λ)

A *Poisson random variable* $X \sim \text{Poisson}(\lambda)$ takes nonnegative integer values with PMF

$$p_X(k) \;=\; e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The parameter $\lambda > 0$ is the average rate. The Poisson is the only standard distribution where mean and variance *both* equal λ.

Connection to Binomial. If we split a long window into n tiny sub-windows, each with success probability λ/n, the count of successes is $\text{Bin}(n, \lambda/n)$. As $n \to \infty$, this converges to $\text{Poisson}(\lambda)$. The Poisson is the limiting count of many rare independent events, which is why it shows up wherever you count phone calls, mutations, or typos.

A worked numeric. A help desk receives an average of $\lambda = 12$ calls per

hour, or 1 call per 5-minute window. The probability of *no* calls in the next 5 minutes is $e^{-1} \approx 0.368$, and the probability of exactly one call is also $e^{-1} \approx 0.368$.

Continuous: the probability density function

The PMF idea breaks down for continuous random variables. Suppose X is the height of a randomly chosen adult, measured in metres. What is $\mathbb{P}(X = 1.732)$? The answer is exactly 0: there are uncountably many possible heights, and no individual real number can carry positive probability without the total exceeding 1.

THE DENSITY PARADOX

"What is $\mathbb{P}(X = 1.732\,m)$ if X is a height?"

$0.$

This is not a flaw of the model. There are uncountably many possible heights; no single one can carry positive weight.

The fix is to replace point probabilities with a *density*.

Definition 4: Probability density function (PDF)

A continuous random variable X has *probability density function* $f_X : \mathbb{R} \to [0, \infty)$ if

$$\mathbb{P}(X \in A) \;=\; \int_A f_X(x)\,dx \qquad \text{for every interval } A \subseteq \mathbb{R}.$$

A PDF satisfies $f_X(x) \geq 0$ for every x and $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$.

Intuition

A PDF is *not* a probability. It is a density, probability per unit of x. To turn it into a probability, multiply by dx and integrate over an interval. The total area under the curve is always 1.

DENSITY, NOT PROBABILITY

$f_X(x)$ can exceed 1. What must be at most 1 is the area under the curve over any interval.

The cumulative distribution function: a unifying object

The PMF and PDF are different kinds of objects, applicable to different kinds of random variables. Is there a single function that handles both? Yes. It is the cumulative distribution function.

Definition 5: Cumulative distribution function (CDF)

The *cumulative distribution function* of a random variable X is the function $F_X : \mathbb{R} \to [0,1]$ defined by

$$F_X(x) \;=\; \mathbb{P}(X \le x).$$

The CDF is non-decreasing, satisfies $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to\infty} F_X(x) = 1$, and is right-continuous.

The CDF is defined for *every* random variable, discrete or continuous. The PMF and PDF are recoverable from it.

Proposition 1: Recovering the PMF / PDF from the CDF

Let X have CDF F_X.

- *If X is discrete, the PMF is the jump heights of the CDF: $p_X(x) = F_X(x) - F_X(x^-)$, where $F_X(x^-) = \lim_{y\uparrow x} F_X(y)$.*
- *If X is continuous and F_X is differentiable at x, the PDF is the derivative: $f_X(x) = F_X'(x)$.*

Sketch. For discrete X, the event $\{X \le x\}$ minus the event $\{X < x\}$ is exactly $\{X = x\}$, so the right-side jump of F_X at x has height $\mathbb{P}(X = x) = p_X(x)$. For continuous X, the fundamental theorem of calculus gives $\frac{d}{dx}\int_{-\infty}^{x} f_X(t)\,dt = f_X(x)$. ■

To make the recovery concrete, take the Bernoulli(p) from earlier: it takes the value 0 with probability $1-p$ and the value 1 with probability p. Its CDF is

$$F_X(x) \;=\; \begin{cases} 0, & x < 0, \\ 1-p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

A staircase with two jumps. The first (at $x = 0$) has height $1-p$, exactly $p_X(0)$; the second (at $x = 1$) has height p, exactly $p_X(1)$. The PMF really is the jump heights of the CDF, as **??** 1 promises.

The standard continuous distributions

Three continuous distributions show up repeatedly in the rest of the book: the Uniform, the Exponential, and the Normal. Each comes with a short story, its PDF, a derivation of its CDF (which we can now do thanks to the definition above), and a numerical example. As with the discrete cases, the mean and variance are stated as facts; the integral techniques behind them belong to Section 2.3.

● UNIFORM IN THE WILD

Where every value in an interval is equally likely: the angle of a spun roulette wheel; the output of a software random-number generator; the position of a dart thrown at random on a board; the time of arrival of a uniformly-distributed signal.

The simplest continuous distribution gives every point in an interval equal weight. ●

Example 7: Uniform(a, b)

The *uniform* random variable $X \sim \text{Uniform}(a, b)$ has PDF

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

For any subinterval $[c, d] \subseteq [a, b]$, $\mathbb{P}(c \le X \le d) = (d - c)/(b - a)$.
CDF. For $a \le x \le b$,

$$F_X(x) = \int_a^x \frac{1}{b-a}\, dt = \frac{x-a}{b-a},$$

linearly ramping from 0 at $x = a$ to 1 at $x = b$ (with $F_X(x) = 0$ for $x < a$ and $F_X(x) = 1$ for $x > b$).
Mean: $(a + b)/2$, the midpoint, exactly as intuition demands. Variance: $(b - a)^2/12$.

If events arrive "out of nowhere" at a constant rate, the waiting time between them is exponentially distributed. ●

● EXPONENTIAL IN THE WILD

The waiting time until something rare happens: time until the next radioactive decay event; time between earthquakes on a fault line; lifespan of a lightbulb (or any component without ageing); time between visits to a low-traffic web page; inter-arrival times of customers at a quiet shop.

Example 8: Exponential(λ)

An *Exponential random variable* $X \sim \mathrm{Exp}(\lambda)$ with rate $\lambda > 0$ has PDF

$$f_X(x) \;=\; \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

CDF. For $x \geq 0$,

$$F_X(x) = \int_0^x \lambda e^{-\lambda t}\, dt = 1 - e^{-\lambda x},$$

and $F_X(x) = 0$ for $x < 0$. Mean: $1/\lambda$. Variance: $1/\lambda^2$. Mean and standard deviation both equal $1/\lambda$.

The Exponential has one celebrated structural property worth deriving here: it is *memoryless*.

Proposition 2: Memoryless property of the Exponential

If $X \sim \mathrm{Exp}(\lambda)$, then for all $s, t \geq 0$,

$$\mathbb{P}(X > s + t \mid X > s) \;=\; \mathbb{P}(X > t).$$

Proof. $\mathbb{P}(X > x) = 1 - F_X(x) = e^{-\lambda x}$ for $x \geq 0$. So

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t). \qquad \blacksquare$$

Intuition

A lightbulb that has already burned for a year, if its lifetime is exponential, is statistically indistinguishable from a brand-new bulb. This is why exponentials model component lifetimes *when there is no ageing*: a radioactive nucleus, an idealised electronic component, or a customer arrival in a queue with no priority.

The most important distribution in the book.• Almost every limit theorem in Parts I, IV, and V converges to a Normal in some way.

• NORMAL IN THE WILD

Almost any quantity that is the sum of many small independent contributions: heights and weights in a population; measurement errors from independent sources; test scores in a large class; sample averages from any distribution with finite variance (the central limit theorem, Chapter 5).

Example 9: Normal $\mathcal{N}(\mu, \sigma^2)$

The *Normal* (or *Gaussian*) random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ has PDF

$$f_X(x) \;=\; \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

The bell curve, centred at μ with spread controlled by $\sigma > 0$. Mean: μ. Variance: σ^2.
CDF. The Normal CDF has no elementary closed form. We write it as

$$F_X(x) \;=\; \Phi\left(\frac{x-\mu}{\sigma}\right),$$

where Φ is the CDF of the *standard* Normal $\mathcal{N}(0,1)$. Tables, software, and Python's `scipy.stats.norm.cdf` all evaluate Φ numerically.
The empirical rule. For $X \sim \mathcal{N}(\mu, \sigma^2)$, the probability that X lies within one, two, or three standard deviations of μ is approximately 0.68, 0.95, and 0.997 respectively. These are the canonical "rules of thumb" for thinking with Normals.

The Normal will reappear repeatedly: in Chapter 3 we compute its moment generating function, in Chapter 5 we prove that sample averages from *any* distribution with finite variance converge to a Normal, and in Parts III and IV the Normal becomes the building block of multivariate Gaussian random vectors and Wishart-distributed sample covariance matrices.

Figure 2.3 plots the three continuous densities side by side at standard parameter values, and Figure 2.4 pairs that with the corresponding CDFs (with the discrete Bernoulli CDF added for contrast). The discrete staircase next to the continuous shapes makes the unifying role of the CDF visible at a glance.

A reference table of standard distributions

The book leans on the same handful of standard distributions repeatedly. Table 2.1 gathers them in one place; later chapters point back to this table rather than re-defining each distribution.

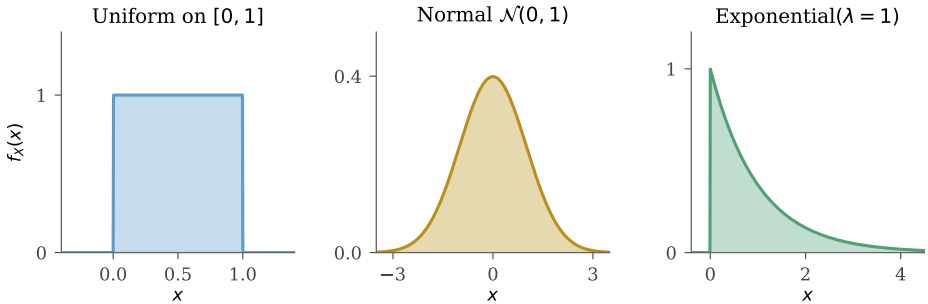


Figure 2.3: Three densities at standard parameter values. The Uniform on $[0,1]$ is constant on its support; the Normal $\mathcal{N}(0,1)$ is the bell curve centred at zero; the Exponential(1) decays from a finite peak at zero. Areas under all three curves are 1.

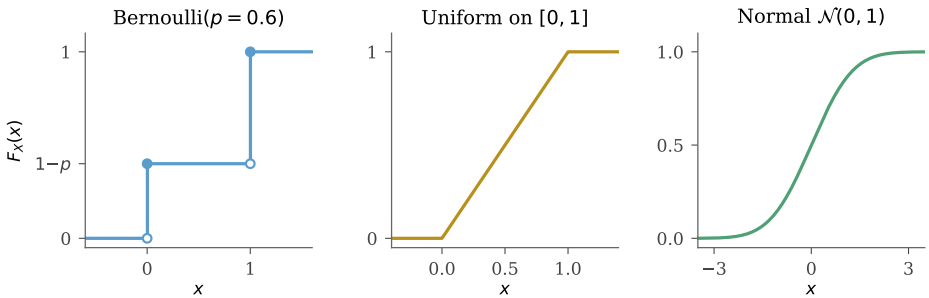


Figure 2.4: The CDF of a discrete random variable is a staircase (Bernoulli(0.6), left); the CDF of a uniformly distributed random variable is a linear ramp (Uniform on $[0,1]$, middle); the CDF of a Normal is a smooth S-curve (right). All three are non-decreasing and approach 1 as $x \to \infty$.

Name	Type	Support	PMF / PDF	Mean	Variance
Bernoulli(p)	discrete	$\{0,1\}$	$p_X(1) = p,\ p_X(0) = 1-p$	p	$p(1-p)$
Binomial(n,p)	discrete	$\{0,1,\ldots,n\}$	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	discrete	$\{1,2,3,\ldots\}$	$(1-p)^{k-1}p$	$1/p$	$(1-p)/p^2$
Poisson(λ)	discrete	$\{0,1,2,\ldots\}$	$e^{-\lambda}\lambda^k/k!$	λ	λ
Uniform(a,b)	continuous	$[a,b]$	$1/(b-a)$ on $[a,b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	continuous	$[0,\infty)$	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$
Normal(μ,σ^2)	continuous	$\mathbb{R}$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

Table 2.1. *The standard distributions used throughout this book. Mean and variance are derived above (and re-derived more cleanly via moment generating functions in Chapter 3).*

The distribution sandbox

● DISTRIBUTION SANDBOX
COLAB: OPEN NOTEBOOK

Reading a PDF or PMF formula on a page is one thing ●; watching a simulator throw thousands of samples and seeing the histogram resolve into the theoretical shape is another. Figure 2.5 shows a snapshot from the companion Colab notebook: 5,000 samples from $\mathrm{Bin}(20, 0.3)$ on the left and from $\mathcal{N}(0, 1)$ on the right, each plotted against the theoretical PMF or PDF.

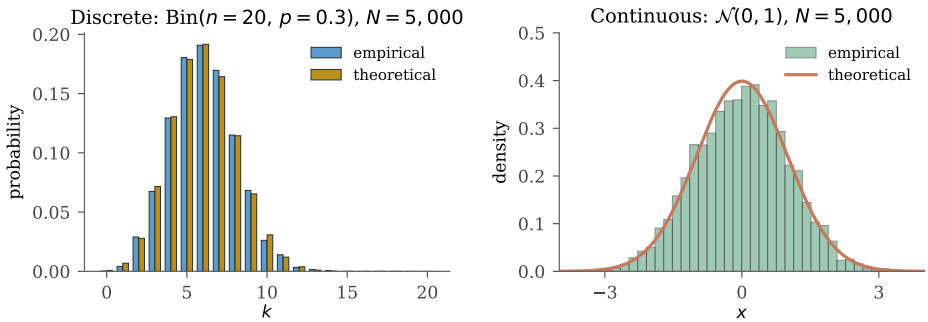


Figure 2.5: Empirical histogram (left bars / shaded) overlaid with the theoretical PMF / PDF (right bars / smooth curve) for two distributions, $N = 5{,}000$ samples each. With this many samples the two are visually indistinguishable; with $N = 50$ they would not be.

The companion notebook `ch01-distribution-sandbox.ipynb` runs the same simulation interactively and lets the reader vary N, the sample size, to watch the empirical bars converge to the theoretical curves. The convergence is one form of the *law of large numbers*, which we prove in Chapter 4.

Brief warm-ups

Hint. ● Use the Bernoulli mean and variance formulas with $p = 0.3$.

Warm-up 2.4 ●. *Let* $X \sim \mathrm{Bernoulli}(0.3)$. ● *Compute* $\mathbb{P}(X = 1)$, $\mathbb{E}[X]$, *and* $\mathrm{Var}(X)$.

► **Reveal answer**

Hint. ● A true-false quiz with random guessing has $p = 1/2$ per question. Use the Binomial mean.

Warm-up 2.5 ●. *A student guesses randomly on a* 10*-question true-false quiz.* ● *Let* X *be the number of correct answers. What is the expected number correct, and what is* $\mathbb{P}(X = 10)$*?*

▶ **Reveal answer**

Hint. ● Geometric mean is $1/p$; here $p = 1/6$.

Warm-up 2.6 ●. *You roll a fair six-sided die until you get a six.* ● *What is the expected number of rolls?*

▶ **Reveal answer**

Hint. ● Convert the rate to "per 5 minutes," then evaluate the Poisson PMF at $k = 2$.

Warm-up 2.7 ●●. *A help desk receives an average of $\lambda = 12$ calls per hour.* ● *Treating the hourly count as Poisson, what is the probability of* exactly two *calls in the next* 5 *minutes?*

▶ **Reveal answer**

Hint. ● Probability over an interval is its length, scaled by $1/(b-a)$.

Warm-up 2.8 ●. *For $X \sim \mathrm{Uniform}(0, 10)$,* ● *compute $\mathbb{P}(3 \le X \le 7)$ and $\mathbb{E}[X]$.*

▶ **Reveal answer**

Hint. ● Memoryless: $\mathbb{P}(X > 2 + 1 \mid X > 2) = \mathbb{P}(X > 1)$.

Warm-up 2.9 ●●. *For $X \sim \mathrm{Exp}(\lambda)$,* ● *verify that $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$ for any $s, t \ge 0$.*

▶ **Reveal answer**

Hint. ● The empirical rule says $\mathbb{P}(|X - \mu| \le \sigma) \approx 0.68$.

Warm-up 2.10 ••. *Adult heights in a population are approximately* $\mathcal{N}(170, 10^2)$ *(centimetres).* ● *Roughly what fraction of the population is between* 160 *and* 180 *cm tall?*

► **Reveal answer**

Warm-up 2.11 ••. *Suppose* X *is continuous with PDF* $f_X(x) = cx$ *for* $x \in [0, 2]$*, and* $f_X(x) = 0$ *otherwise.* ● *Find the constant* c *and compute* $\mathbb{P}(X \leq 1)$ *(equivalently, the value of* $F_X(1)$*).*

► **Reveal answer**

2.3 Expectation and variance

Placeholder content for this section. §1.3: definition and intuition; linearity of expectation; variance and standard deviation; worked examples on the running distributions.

2.4 Independence and joint distributions

Placeholder content for this section. §1.4: two random variables at once; joint and marginal distributions; independence and its consequences; second Colab simulation (joint plots for independent RVs).

2.5 *Brief warm-ups and chapter recap*

Placeholder content for this section. §1.5: 3–5 cross-section warm-ups; chapter recap box.

Chapter recap

- *Placeholder content for this section. Punchline 1.*
- *Placeholder content for this section. Punchline 2.*
- *Placeholder content for this section. Punchline 3.*

Exercises

Placeholder content for this section. 8–15 longer exercises; solutions in Appendix A.

3

Generating Functions

Where we are

We've defined random variables and their distributions. Now we ask: what would it look like to package an entire distribution into a single, easy-to-manipulate object? Two such packages turn out to be central in everything that follows: the moment generating function (MGF) and the characteristic function (CF).

Placeholder content for this section. This chapter has the most mature source material in the existing notes. Pull from `Mathematics_Research_Notes/main.tex`*, the MGF and CF sections (lines 13–585). Use as the canonical reference for visual style of every other chapter.*

3.1 *Why bundle a distribution into a function?*

Placeholder content for this section. Motivation: sums of independent RVs become products; moments become derivatives; Laplace/Fourier transform connections.

3.2 *The moment generating function*

Placeholder content for this section. Definition, domain, MGF as a two-sided Laplace transform, MGF generates moments, examples (Bernoulli, Binomial, Poisson, Exponential, Normal).

3.3 *Sums of independent random variables*

Placeholder content for this section. Multiplicativity; Poisson-plus-Poisson example; Normal-plus-Normal example.

3.4 *When the MGF fails: characteristic functions*

Placeholder content for this section. Cauchy distribution as motivation; CF always exists; relationship $\varphi_X(t) = M_X(it)$ *when both exist.*

3.5 *Characteristic function as Fourier transform*

Placeholder content for this section. Fourier and Fourier–Stieltjes viewpoint.

3.6 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions with immediate answers.

Chapter recap

- *Placeholder content for this section. MGFs encode distributions analytically and turn sums into products.*
- *Placeholder content for this section. CFs always exist; MGFs sometimes don't.*
- *Placeholder content for this section. Both will be the engine of the CLT in Chapter 4.*

Exercises

Placeholder content for this section. 8–15 longer exercises; solutions in Appendix A.

4

Convergence: How Sequences of Random Things Settle Down

Where we are

A single random variable is one thing; a *sequence* of random variables is the entry point into asymptotic statistics. Three different notions of "settling down" coexist, almost surely, in probability, in distribution, with a clean implication chain between them.

Placeholder content for this section. Sources: existing main.tex 'Modes of Convergence' section; Joseph thesis Section 2.2; Casella & Berger Section 5.5.

4.1 Almost sure convergence

Placeholder content for this section. Definition, sample-path interpretation, examples (shifted RVs, $X + 1/n$).

4.2 Convergence in probability

Placeholder content for this section. Definition, contrast with a.s. the canonical Borel–Cantelli counterexample.

4.3 Markov's and Chebyshev's inequalities

Placeholder content for this section. Statements, proofs, used in the WLLN.

4.4 The Weak Law of Large Numbers

Placeholder content for this section. Statement, Chebyshev-based proof.

4.5 Convergence in distribution

Placeholder content for this section. Definition via CDFs, contrast with the other modes, why it's the right notion for the CLT.

4.6 The implication chain

Placeholder content for this section. a.s. $\Rightarrow$ in prob $\Rightarrow$ in dist; converses fail.

4.7 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions with immediate answers.

Chapter recap

- *Placeholder content for this section. Three modes, ordered from strongest to weakest.*
- *Placeholder content for this section. Markov + Chebyshev are the technical workhorses.*
- *Placeholder content for this section. Convergence in distribution is what limit theorems are about.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

5

The Central Limit Theorem

Where we are

The capstone of Part I. Lévy's continuity theorem hands us a tool to convert any pointwise statement about characteristic functions into a statement about convergence in distribution. We use it to prove the CLT, which sits behind essentially every statement in the rest of the book.

Placeholder content for this section. Sources: existing main.tex 'Characteristic Functions and Convergence in Distribution' section; the proof is already written there.

5.1 Lévy's continuity theorem

Placeholder content for this section. Statement, intuition, proof sketch (full proof can go to chapter appendix).

5.2 *The classical CLT, via characteristic functions*

Placeholder content for this section. Statement, Taylor expansion of the CF, $\sqrt{n}$ scaling explained, conclusion via Lévy.

5.3 *Why $\sqrt{n}$? Where does the $\frac{1}{2}$ come from?*

Placeholder content for this section. Intuition box; the second moment is the first nonzero term in the CF expansion.

5.4 *The CLT in pictures*

Placeholder content for this section. Histogram of standardized means converging to the bell curve; QQ-plot version; small Colab simulation.

5.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. The CLT is a statement about $\sqrt{n}$-scaled fluctuations.*
- *Placeholder content for this section. Lévy's theorem is the bridge between CFs and distributional convergence.*
- *Placeholder content for this section. This Gaussian universality is one of two big universality phenomena in this book; the other, Tracy–Widom, waits in Part V.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Part II

From Numbers to Vectors and Matrices

Random matrix theory only makes sense once you stop thinking about a single random number and start thinking about a whole vector, or even a whole matrix, of them. Part II re-tools the foundations to handle vectors and the matrices that summarize their joint behaviour.

Chapters in this part: 5. Random Vectors 6. A Linear Algebra Refresher 7. Covariance Matrices.

6

Random Vectors and Joint Distributions

Where we are

Random matrices are made of random numbers, but more directly, of random *vectors* arranged in rows or columns. Before we can study random matrices, we have to understand what it means for an entire vector of random numbers to have a distribution.

Placeholder content for this section. Sources: existing main.tex 'Random Vectors' subsection; Joseph thesis Sections 2.4 (Defs 2.11–2.14, Prop 2.3, Example 2.10).

6.1 *What is a random vector?*

Placeholder content for this section. Definition; joint CDF as a map $\mathbb{R}^p \to [0,1]$*; key properties; marginals via limits.*

6.2 *Independence in higher dimensions*

Placeholder content for this section. Joint CDF factors; example (independent exponentials); contrast with the maximally dependent case ($X_1 = X_2$ *a.s.).*

6.3 *The multivariate normal distribution*

Placeholder content for this section. Density formula; bivariate normal worked out; the special Gaussian fact "zero covariance implies independence," valid only in the Gaussian family.

6.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. A random vector packages multiple random variables with their joint distribution.*
- *Placeholder content for this section. Multivariate normality is the workhorse joint distribution and has special independence properties.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

7

A Linear Algebra Refresher

Where we are

We need eigenvalues and eigenvectors, and we need them *understood*, not just defined. This chapter rebuilds the linear algebra of symmetric matrices from scratch: eigenvalues, the spectral theorem, matrix norms, and the equivalence of all norms in finite dimensions.

Placeholder content for this section. Sources: existing main.tex spectral-theorem subsection; Gilbert Strang Introduction to Linear Algebra *for accessible-textbook tone.*

7.1 *Eigenvalues, eigenvectors, the spectral theorem*

Placeholder content for this section. From scratch: $Av = \lambda v$*, geometric meaning, characteristic polynomial; the spectral theorem* $A = Q\Lambda Q^T$ *for real symmetric* A*.*

7.2 *Quadratic forms and what eigenvalues control*

Placeholder content for this section. The diagonal form $x^T Ax = \sum \lambda_i y_i^2$; sign of eigenvalues controls definiteness.

7.3 *Matrix norms*

Placeholder content for this section. Frobenius and operator norms; submultiplicativity; equivalence of norms in finite dimensions.

7.4 *Continuity of eigenvalues*

Placeholder content for this section. Determinant is a polynomial in entries; characteristic polynomial coefficients are continuous; roots of polynomials depend continuously on coefficients; therefore eigenvalues depend continuously on matrix entries. This is the analytic engine of Chapter 9.

7.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Spectral theorem: real symmetric $\Leftrightarrow$ rotation-scale-rotation.*
- *Placeholder content for this section. All norms equivalent in finite dimensions.*
- *Placeholder content for this section. Eigenvalues are continuous functions of the matrix.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

8

Covariance Matrices and Positive (Semi-)Definiteness

Where we are

We package the joint behaviour of a random vector into a single matrix, the *covariance matrix*, and meet the linear-algebraic constraint that all such matrices share: positive semi-definiteness. This is the algebraic gate keeping our matrices inside a well-behaved corner of matrix space.

Placeholder content for this section. Sources: existing main.tex 'Covariance and Covariance Matrices' subsection.

8.1 Covariance and the covariance matrix

Placeholder content for this section. Definition; entrywise structure; symmetry; $p = 2$ explicit form.

8.2 Why covariance matrices are PSD

Placeholder content for this section. Proof via $v^T \Sigma v = \mathrm{Var}(v^T X) \geq 0$.

8.3 When is a covariance matrix PD?

Placeholder content for this section. Σ fails to be PD iff some nontrivial linear combination collapses to a constant.

8.4 Eigenvalue characterization

Placeholder content for this section. Σ PSD iff all eigenvalues ≥ 0; PD iff strictly > 0. Connect to Chapter 6.

8.5 Brief warm-ups

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Σ encodes the joint covariance structure as a symmetric PSD matrix.*
- *Placeholder content for this section. Eigenvalues ≥ 0 characterize PSD.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Part III

Sample Covariance Matrices

In real life, we never know the population covariance, we estimate it from a finite sample. The sample covariance matrix *is, on the surface, just a calculation; on closer inspection it is a structured random matrix whose entire spectrum carries statistical information. Part III treats it as such.*

Chapters in this part: 8. The Sample Covariance Matrix 9. Eigenvalue Convergence 10. The Wishart Distribution.

9

The Sample Covariance Matrix

Where we are

In real life Σ is unknown; we estimate it from data. The estimator is the *sample covariance matrix*, which on the surface is just an arithmetic average of outer products, but on closer inspection is a structured random matrix whose spectrum carries statistical information.

Placeholder content for this section. Sources: existing main.tex 'Sample Covariance Matrix' subsection.

9.1 From population to sample

Placeholder content for this section. Sample mean vector; mean-known-zero case ($\widehat{\Sigma}_n = \frac{1}{n} X^T X$); mean-unknown case ($\frac{1}{n-1}$ centered).

9.2 Numerical examples

Placeholder content for this section. Toy 2D example with three observations; perfect-correlation example; mean-zero example.

9.3 $\widehat{\Sigma}_n$ as a random matrix

Placeholder content for this section. Symmetry; PSD; rank under i.i.d. Gaussian assumption (full rank when $n \geq p$).

9.4 Brief warm-ups

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. $\widehat{\Sigma}_n$ estimates Σ from finite data.*
- *Placeholder content for this section. $\widehat{\Sigma}_n$ inherits symmetry and PSD from Σ.*
- *Placeholder content for this section. Treating $\widehat{\Sigma}_n$ as a random matrix is the gateway to the rest of this book.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

10

Eigenvalues Are Continuous Functions of the Matrix

Where we are

This is Joseph's Proposition 3.1: when matrices converge, their eigenvalues converge. In the deterministic, almost-sure, and in-probability flavours, the same continuity argument carries the day. This is the spine on which every later eigenvalue convergence claim hangs.

Placeholder content for this section. Sources: existing main.tex 'Continuation: Sample Covariance Matrices, Norms, and Eigenvalue Convergence' section.

10.1 *The logical chain*

Placeholder content for this section. Determinant continuous (polynomial in entries) $\to$ characteristic polynomial coefficients continuous $\to$ roots continuous $\to$ eigenvalues continuous.

10.2 *Three convergence flavours*

Placeholder content for this section. Deterministic, a.s., in probability. Each follows from the deterministic version applied either pointwise on a measure-1 set or via continuity bounds.

10.3 *Sanity check: the 2×2 case*

Placeholder content for this section. Explicit eigenvalue formula in 2D; entrywise convergence trivially gives eigenvalue convergence.

10.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Matrix convergence implies eigenvalue convergence.*
- *Placeholder content for this section. The pipeline is: entries $\to$ coefficients $\to$ roots.*
- *Placeholder content for this section. This is the analytic engine for the next several chapters.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

11

The Wishart Distribution

Where we are

When the data is Gaussian, the unscaled sample covariance matrix has a name: the *Wishart distribution*. It is the multivariate generalization of the chi-square law, and it is the cleanest analytic object in this book.

Placeholder content for this section. Sources: existing main.tex Wishart sections (continuation Apr 16); Joseph thesis Section 3.2 Def 3.3.

11.1 Definition and density

Placeholder content for this section. $S = \sum X_i X_i^T$ *for Gaussian* X_i*; the Wishart density formula.*

11.2 Dimension one collapses to chi-square

Placeholder content for this section. Worked specialization. Why this is natural from the construction.

11.3 *Dimension two: structure of the Wishart matrix*

Placeholder content for this section. Diagonal χ^2_n, off-diagonal $\sum x_i y_i$, basic moment computations.

11.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Wishart is the multivariate chi-square.*
- *Placeholder content for this section. In dimension 1 it* is *chi-square.*
- *Placeholder content for this section. In dimension 2 it has clean structure that powers Chapters 11–14.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Part IV

Eigenvalue Fluctuations in Fixed Dimension

The sample covariance matrix converges to the population covariance matrix. But how *does it fluctuate around the limit? The answer depends on whether the population eigenvalues are repeated or distinct, and the two cases lead to fundamentally different limit laws.*

Chapters in this part: 11. Joint CLT for Entries 12. Two Fluctuation Regimes 13. Eigenvector Convergence 14. The GOE.

12

Joint Asymptotic Distribution of the Matrix Entries

Where we are

We zoom from the matrix to its entries. The three distinct entries of $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$ in dimension 2 are jointly asymptotically normal, and we'll compute the exact limiting covariance, including using Isserlis's theorem to verify that the off-diagonal asymptotic covariances vanish.

Placeholder content for this section. Sources: existing main.tex April 16 + April 17 entry-fluctuation sections.

12.1 *The three entries and their marginal CLTs*

Placeholder content for this section. Variances 2, 2, 1 *in the* $\Sigma = I_2$ *case;* 2, 8, 2 *in the* $\Sigma = \mathrm{diag}(1, 2)$ *case.*

12.2 *Isserlis's theorem and the diagonal limit*

Placeholder content for this section. Wick pairings; off-diagonal covariances vanish; limit is diagonal Gaussian.

12.3 *Both cases side by side*

Placeholder content for this section. Case $\Sigma = I_2$ vs $\Sigma = \mathrm{diag}(1, 2)$; same engine, different numbers.

12.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Joint CLT for the three distinct entries.*
- *Placeholder content for this section. Isserlis kills off-diagonal covariances by parity.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

13

Distinct vs Repeated Eigenvalues: Two Fluctuation Regimes

Where we are

The first really surprising fact in the book. The same scaling, $\sqrt{n}$, gives a Gaussian limit when the population eigenvalues are distinct, but a non-Gaussian (GOE-type) limit when they coincide. The reason is that the eigenvalue map is differentiable in one case and only Lipschitz in the other.

Placeholder content for this section. Sources: existing main.tex 'Continuation: Wishart Matrices...' section + April 17 derivative-of-eigenvalue work.

13.1 *Case 1: $\Sigma = I_2$ (repeated eigenvalues)*

Placeholder content for this section. No Taylor expansion needed; eigenvalues of $\widehat{\Sigma}_n$ are eigenvalues of $W_n/\sqrt{n}$ shifted by 1; limit law is non-Gaussian.

13.2 *Case 2:* $\Sigma = \mathrm{diag}(1, 2)$ *(distinct eigenvalues)*

Placeholder content for this section. First-order perturbation theory; $d\lambda = u^T(dM)u$*; delta method; Gaussian limit with variance 8 (top eigenvalue) and 2 (bottom eigenvalue).*

13.3 *The derivative-of-eigenvalue calculation explicitly*

Placeholder content for this section. Three partial derivatives: $\partial\lambda_2/\partial m_{11}, \partial m_{12}, \partial m_{22}$*; gradient* $(0, 0, 1)$ *at* $\Sigma = \mathrm{diag}(1, 2)$*.*

13.4 *Why the same* $\sqrt{n}$ *gives two different laws*

Placeholder content for this section. The matrix scaling is universal; the eigenvalue map is what differs. Map smooth $\Rightarrow$ *delta method; map only Lipschitz* $\Rightarrow$ *nonlinear limit.*

13.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Distinct eigenvalues → Gaussian fluctuations via the delta method.*
- *Placeholder content for this section. Repeated eigenvalues → GOE-like fluctuations.*
- *Placeholder content for this section. Same scaling, two universality classes.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

14

Eigenvectors: When They Converge and When They Don't

Where we are

Convergence of eigenvalues does not imply convergence of eigenvectors. When the limiting eigenvalue is simple, the spectral projector is continuous and a sign convention pins down a converging eigenvector. When eigenvalues coincide, the eigenspace is multi-dimensional and the eigenvectors can rotate freely without changing the matrix limit.

Placeholder content for this section. Sources: April 17 Gustavo notes (rotation counterexample). Existing main.tex April 17 continuation section.

14.1 *Spectral projectors and the continuity argument*

Placeholder content for this section. Riesz contour formula; isolated eigenvalues give continuous projectors; sign convention.

14.2 *When eigenvalues coincide*

Placeholder content for this section. Eigenspace = whole plane; any basis is an eigenbasis; no canonical limit.

14.3 *The rotation counterexample*

Placeholder content for this section. $S_n = O_n \mathrm{diag}(\lambda_1^{(n)}, \lambda_2^{(n)}) O_n^T$ *with* $\theta_n = n\pi/2$*;* $S_n \to I_2$ *in operator norm but the eigenvectors rotate.*

14.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Eigenvector convergence requires eigenvalue isolation.*
- *Placeholder content for this section. The rotation counterexample makes the failure tangible.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

15

The Gaussian Orthogonal Ensemble (GOE)

Where we are

The non-Gaussian limit law from Chapter 12 has a name: the eigenvalue distribution of the *Gaussian Orthogonal Ensemble*. This chapter formalizes the GOE, derives its joint eigenvalue density, and explains the Vandermonde repulsion factor that makes coincident eigenvalues vanishingly unlikely.

Placeholder content for this section. Sources: existing main.tex GOE section; Joseph thesis Section 3.2.

15.1 *Definition of the GOE*

Placeholder content for this section. $N \times N$ *symmetric, diagonal entries* $\mathcal{N}(0,2)$*, off-diagonal* $\mathcal{N}(0,1)$*, all independent up to symmetry. Construction* $H = (G + G^T)/\sqrt{2}$ *from i.i.d. Gaussian* G*.*

15.2 *Joint eigenvalue density and the Vandermonde*

Placeholder content for this section. Density formula with $\prod_{i<j} |\lambda_i - \lambda_j|^{\beta=1}$ *factor; explicit* 2×2 *case.*

15.3 *Eigenvalue repulsion*

Placeholder content for this section. The Vandermonde forces the density to vanish on the diagonal $\lambda_1 = \lambda_2$*; eigenvalues "avoid" each other.*

15.4 *Why GOE specifically appears in our story*

Placeholder content for this section. $\sqrt{n}(\widehat{\Sigma}_n - I_p)$ *converges to a GOE-like matrix in fixed dimension. Closes the loop with Chapter 12.*

15.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. GOE = $\beta = 1$ symmetric Gaussian ensemble.*
- *Placeholder content for this section. Vandermonde factor encodes repulsion.*
- *Placeholder content for this section. Limit law for $\sqrt{n}(\widehat{\Sigma}_n - I_p)$ in fixed p is GOE-shaped.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Part V

High-Dimensional Phenomena

Once the dimension p grows together with the sample size n, the spectral picture changes completely. The empirical eigenvalue distribution stops collapsing to a single point and instead converges to the Marchenko–Pastur *law. The largest eigenvalue lives at the edge of the bulk, with fluctuations governed by the* Tracy–Widom *law.*

Chapters in this part: 15. From Fixed-p to High Dimensions 16. The Marchenko–Pastur Law 17. Tracy–Widom 18. Q–Q Plots.

16

From Fixed-p to High Dimensions

Where we are

Everything in Part IV assumed p fixed and $n \to \infty$. That regime is fine for classical statistics, but modern data has p comparable to (or much larger than) n. The empirical eigenvalue distribution stops collapsing to a point. This chapter is the gateway to the rest of Part V.

Placeholder content for this section. New chapter; pulls from April 17 notes and Joseph thesis Section 3.2 (Prop 3.8).

16.1 *What changes when $p/n \to c > 0$*

Placeholder content for this section. The empirical spectral distribution becomes a continuous law, not a delta.

16.2 *Three pictures*

Placeholder content for this section. p fixed: Dirac at 1; $p/n \to c \in (0,1)$: Marchenko–Pastur on $[\lambda_-, \lambda_+]$; $p > n$: rank deficiency, atom at 0.

16.3 *The simplest example of rank deficiency:* $p = 2, n = 1$

Placeholder content for this section. $\widehat{\Sigma}_n = X_1 X_1^T$ *has* $\det = 0$*; one eigenvalue is exactly zero.*

16.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. High dimensions = p growing with n.*
- *Placeholder content for this section. ESD has nontrivial support, not just a single point.*
- *Placeholder content for this section. When $p > n$, $\widehat{\Sigma}_n$ is rank-deficient.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

17

The Marchenko–Pastur Law

Where we are

The Marchenko–Pastur law is the high-dimensional analogue of "$\widehat{\Sigma}_n \to I_p$." Instead of every eigenvalue converging to 1, the empirical eigenvalue distribution converges to a continuous law on $[(1-\sqrt{c})^2, (1+\sqrt{c})^2]$. This is the universal eigenvalue picture for sample covariance matrices in high dimensions.

Placeholder content for this section. New chapter; pulls from Joseph thesis Section 3.2 (Prop 3.8); reference: Bai & Silverstein, Vershynin Ch. 7.

17.1 Statement of the Marchenko–Pastur law

Placeholder content for this section. Density formula; support $[\lambda_-, \lambda_+] = [(1-\sqrt{c})^2, (1+\sqrt{c})^2]$; convergence in distribution of the empirical spectral distribution.

17.2 *Three perspectives on why it's true*

Placeholder content for this section. (1) Stieltjes-transform sketch (intuition only); (2) free-probability free convolution; (3) numerical experiment. Pointer to the rigorous proof in chapter appendix.

17.3 *What MP looks like as c varies*

Placeholder content for this section. $c \to 0$*: density concentrates near 1 (recovering fixed-p);* $c = 1$*: hits zero on the left edge;* $c > 1$*: atom at 0, support shifts.*

17.4 *Simulation: MP as N grows*

Placeholder content for this section. Colab box: histogram of $\widehat{\Sigma}_n$ *eigenvalues for* $p = 200, 500, 1000$ *overlaid with theoretical density.*

17.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. ESD of $\widehat{\Sigma}_n$ converges to Marchenko–Pastur in the high-dim regime.*
- *Placeholder content for this section. Support is $[(1-\sqrt{c})^2, (1+\sqrt{c})^2]$.*
- *Placeholder content for this section. Density vanishes like a square root at the edges, the seed of Tracy–Widom.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

18

Tracy–Widom: Fluctuations at the Edge

Where we are

The Marchenko–Pastur law tells us the largest sample eigenvalue stabilizes at $\lambda_+ = (1+\sqrt{c})^2$. But how does it *fluctuate*? The answer is the most famous law in random matrix theory: the Tracy–Widom distribution, which sits at the edge between two regimes and inherits a $p^{-2/3}$ scaling from the square-root vanishing of the MP density.

Placeholder content for this section. Sources: April 17 Gustavo notes (TW heuristics + $p^{-2/3}$ derivation); Joseph thesis Section 3.2 (Prop 3.9); Quanta Magazine article for context.

18.1 Two heuristics: edge of two regimes, square-root density

Placeholder content for this section. (1) TW lives at the boundary between Vandermonde-repelled bulk and empty exterior. (2) MP density $\sim C(\lambda_+ - \lambda)^{1/2}$ near the edge.

18.2 Heuristic derivation of the $p^{-2/3}$ scaling

Placeholder content for this section. The "1/p mass in the interval $[\lambda_{\max}, \lambda_+]$" argument, integrated against $C(\lambda_+ - \lambda)^{1/2}$, gives $\lambda_+ - \lambda_{\max} \sim p^{-2/3}$.

18.3 Statement of the Tracy–Widom limit

Placeholder content for this section. $p^{2/3}(\lambda_{\max} - \lambda_+) \xrightarrow{d} \mathrm{TW}_1$, with model-dependent prefactor; properties of TW1 (mean, std, skewness).

18.4 Analogy with the CLT

Placeholder content for this section. $\sqrt{n}$-CLT vs $p^{2/3}$-Tracy-Widom: in both cases, the unique scaling that stabilizes the fluctuation.

18.5 Simulation: TW1 from GOE top eigenvalues

Placeholder content for this section. Colab box: simulate top eigenvalue of $N \times N$ GOE for growing N; histogram converges to TW1 shape (left-skewed).

18.6 Brief warm-ups

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. TW1 is the limiting law of $\lambda_{\max}$ at the spectral edge.*
- *Placeholder content for this section. $p^{-2/3}$ scaling falls out of the square-root vanishing of MP.*
- *Placeholder content for this section. TW1 is to the spectral edge what the Gaussian is to the sample mean.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

19

Q–Q Plots as Diagnostics

Where we are

A Q–Q plot is the workhorse visual tool for asking "does this sample look Gaussian?", and, more interestingly, "does this sample look *not* Gaussian?" This chapter builds Q–Q plots from scratch, applies them to the limit laws of Parts IV and V, and uses them to see TW1 differ from the Gaussian.

Placeholder content for this section. Sources: existing main.tex Q–Q plot section; the existing Colab notebook on TW1 vs Gaussian QQ.

19.1 What a Q–Q plot is and isn't

Placeholder content for this section. Theoretical quantiles vs sample quantiles; not *a CDF plot; reading shape (heavy/light tails, scale, skew).*

19.2 Three baselines: Gaussian, Cauchy, exponential means

Placeholder content for this section. Gaussian sample vs Gaussian quantile (line); Cauchy sample (heavy tails); standardized exponential means (CLT in pictures).

19.3 Q–Q for sample covariance eigenvalues

Placeholder content for this section. Distinct vs repeated case revisited; Q–Q plots distinguish the two regimes visually.

19.4 Q–Q for Tracy–Widom vs Gaussian

Placeholder content for this section. Standardized TW1 sample vs $\mathcal{N}(0,1)$ *quantiles; left-tail dip, right-tail flatness; the asymmetry signature.*

19.5 Simulation: full Colab notebook

Placeholder content for this section. End-to-end Colab box: GOE simulation, normalization, QQ plot, residual plot, scaling sweep with N*.*

19.6 Brief warm-ups

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Q–Q plots are quantile-vs-quantile, not CDF-vs-CDF.*
- *Placeholder content for this section. Departures from $y = x$ encode tail behaviour and skewness.*
- *Placeholder content for this section. TW1 vs Gaussian QQ shows the universality class boundary.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Part VI

Spikes and Spectral Clustering

When the population covariance is the identity plus a low-rank perturbation, a few eigenvalues escape the Marchenko–Pastur bulk, if and only if the perturbation is strong enough. This is the BBP phase transition, and it is exactly the mechanism behind spectral clustering*: the destination this entire book has been pointing at.*

Chapters in this part: 19. Spiked Covariance Models 20. The BBP Phase Transition 21. Spectral Clustering: Three Perspectives.

20

Spiked Covariance Models

Where we are

The Marchenko–Pastur picture describes the eigenvalues of a sample covariance matrix when the population is isotropic ($\Sigma = I_p$). But what if the population has *structure*, a low-rank "signal" on top of isotropic noise? That structure either does or does not produce visible eigenvalue spikes above the bulk; this chapter sets up the question and motivates the answer.

Placeholder content for this section. Mostly new chapter. Reference: Yale Stat 615 Lecture 9, Vershynin Ch. 7–8.

20.1 The spiked model: noise plus low-rank signal

Placeholder content for this section. $\Sigma = I_p + \sum_k \theta_k v_k v_k^T$. *PCA motivation. Why this is the canonical model in modern data science.*

20.2 *Examples in the wild*

Placeholder content for this section. Factor models in finance, latent semantic analysis, gene expression PCA, deep learning Hessian.

20.3 *What a spike "looks like" in simulations*

Placeholder content for this section. Histograms with spike clearly above the MP bulk; scatter of (sample top eigenvalue, true spike strength).

20.4 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Spiked model = isotropic noise + structured low-rank signal.*
- *Placeholder content for this section. Spikes show up in the spectrum as outliers above the MP bulk.*
- *Placeholder content for this section. Whether they show up depends on signal strength, BBP transition in Chapter 20.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

21

The BBP Phase Transition

Where we are

The BBP (Baik–Ben Arous–Péché) phase transition is the precise mathematical statement of when a spike escapes the Marchenko–Pastur bulk. There is a critical signal strength below which the spike is invisible (eigenvalue stays at the bulk edge) and above which it becomes detectable (eigenvalue moves outside the bulk). This is a phase transition in the literal mathematical sense.

Placeholder content for this section. Mostly new chapter. Reference: Yale Stat 615 Lecture 9, original BBP paper.

21.1 *Sub-critical and super-critical regimes*

Placeholder content for this section. Critical strength $\theta^* = \sqrt{c}$. Below: spike eigenvalue stays at λ_+. Above: spike eigenvalue at $\theta + c\theta/(\theta - 1)$ or similar formula.

21.2 *Three pictures of the transition*

Placeholder content for this section. Sub-critical histogram, super-critical histogram, θ sweep showing the phase transition.

21.3 *Detectability and PCA*

Placeholder content for this section. Why this matters: PCA is only useful above the BBP threshold. Below threshold, the top sample eigenvector is essentially random.

21.4 *Simulation: BBP transition sweep*

Placeholder content for this section. Colab box: parameter sweep over θ with histogram of top sample eigenvalue.

21.5 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. BBP gives a sharp threshold for spike detection.*
- *Placeholder content for this section. Below threshold: spike invisible.*
- *Placeholder content for this section. Above threshold: spike detectable, eigenvalue moves outside the bulk.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

22

Spectral Clustering: Three Perspectives

Where we are

The destination. *Spectral clustering* is what happens when you take the spiked-covariance picture seriously: the signal eigenvectors of a similarity matrix, graph Laplacian, or kernel reveal cluster structure in the data. We follow von Luxburg and present the algorithm three different ways, so that if one explanation doesn't click, the other two will.

Placeholder content for this section. New chapter. Primary source: von Luxburg's A Tutorial on Spectral Clustering *(the canonical pedagogical reference).*

22.1 *The algorithm in one paragraph*

Placeholder content for this section. Build similarity graph; compute graph Laplacian; take its smallest k eigenvectors; cluster the n embedded points with k-means.

22.2 *Perspective 1: graph cut*

Placeholder content for this section. Normalized cut interpretation; spectral relaxation of an NP-hard problem.

22.3 *Perspective 2: random walk*

Placeholder content for this section. Random walk on the similarity graph; the Laplacian eigenvectors are stationary modes.

22.4 *Perspective 3: perturbation theory*

Placeholder content for this section. The Laplacian of a block-structured graph has a specific spectrum; perturbations don't break the picture too much. This is where the BBP intuition feeds in directly.

22.5 *Why this is the destination of the entire book*

Placeholder content for this section. The full chain: Gaussian foundations $\to$ random matrices $\to$ spiked covariance $\to$ BBP detectability $\to$ clusters revealed by top eigenvectors. Spectral clustering is exactly the algorithmic payoff.

22.6 *Simulation: spectral clustering on a synthetic dataset*

Placeholder content for this section. Colab box: two-blob dataset, build kernel, compute Laplacian eigenvectors, k-means in spectral coordinates.

22.7 *Brief warm-ups*

Placeholder content for this section. 3–5 short questions.

Chapter recap

- *Placeholder content for this section. Spectral clustering = top/bottom Laplacian eigenvectors + k-means.*
- *Placeholder content for this section. Three independent reasons it works.*
- *Placeholder content for this section. Random matrix theory is the explanation for* when *it works.*

Exercises

Placeholder content for this section. 8–15 longer exercises.

Coda: Where to Next

Placeholder content for this section. This closing chapter is a short tour of the territory beyond this book. To be filled in after the main chapters are drafted.

Universality classes beyond Tracy–Widom

Placeholder content for this section. TW is one universality class. There are others (Pearcey, Airy, sine, β-ensembles). Brief tour.

Multiscale phenomena

Placeholder content for this section. Joseph Genzer's honors thesis introduced fractional Brownian motion and a phase transition at $H = 0.75$. Brief teaser of that storyline.

Where this stuff actually shows up

Placeholder content for this section. Connections to deep learning (training dynamics, NTK, Hessian spectra), finance (portfolio risk, factor models), wireless communications (MIMO capacity), and biology (PCA in genomics).

Where to read next

Placeholder content for this section. Annotated reading list: Tao's Topics in Random Matrix Theory, *Vershynin's* High-Dimensional Probability, *Bai and Silverstein, Anderson–Guionnet–Zeitouni, Couillet and Debbah, the Wikipedia Tracy–Widom article (yes, really), Joseph's thesis itself.*

Solutions to End-of-Chapter Exercises

Placeholder content for this section. This appendix collects the full solutions to the end-of-chapter exercises. Section per chapter; entry per exercise. Filled in incrementally as chapters are written.

Solutions for Chapter 1

Placeholder content for this section. To be filled in.

Solutions for Chapter 2

Placeholder content for this section. To be filled in.

Solutions for Chapter 3

Placeholder content for this section. To be filled in.

Notation Index

Placeholder content for this section. The full notation index is here. The frontmatter notation guide is the seed; this appendix is the comprehensive list, sortable by symbol with chapter-of-introduction.

Symbol	**Meaning**	**First used**
$\mathbb{R}, \mathbb{N}, \mathbb{Z}$	real, natural, integer numbers	Ch. 1
$\mathbb{E}[X]$	expectation of X	Ch. 1
$\mathrm{Var}(X)$	variance of X	Ch. 1
$M_X(t)$	moment generating function of X	Ch. 2
$\varphi_X(t)$	characteristic function of X	Ch. 2
$\xrightarrow{\text{a.s.}}, \xrightarrow{P}, \xrightarrow{d}$	convergence (a.s., probability, distribution)	Ch. 3
Σ	population covariance matrix	Ch. 7
$\widehat{\Sigma}_n$	sample covariance matrix	Ch. 8
$\lambda_i(M)$	ith eigenvalue of M	Ch. 6
$\mathcal{W}_p(n, \Sigma)$	Wishart distribution	Ch. 10
f_{MP}	Marchenko–Pastur density	Ch. 16
TW_1	Tracy–Widom type 1	Ch. 17

Placeholder content for this section. Expand as chapters are filled in.

Code Listings and Colab Notebook Index

Placeholder content for this section. This appendix is a directory of every simulation in the book, with: short description, chapter where it appears, link to the live Colab notebook, and (where short enough) the inline Python code. The point is that a reader can fork a notebook for any experiment in the book in one click.

Simulation	**Chapter**	**Topic**	**Notebook**
Sim. 4.1	4	CLT in pictures	*Placeholder content for this section. link*
Sim. 16.1	16	MP as N grows	*Placeholder content for this section. link*
Sim. 17.1	17	TW1 from GOE top eigenvalue	*Placeholder content for this section. link*
Sim. 18.1	18	QQ-plot diagnostic suite	*Placeholder content for this section. link*
Sim. 20.1	20	BBP transition sweep	*Placeholder content for this section. link*
Sim. 21.1	21	Spectral clustering on a synthetic dataset	*Placeholder content for this section. link*

Placeholder content for this section. Expand as Colab notebooks are written. Each Colab box in the body text should have a matching entry here.

Mathematical Prerequisites

Placeholder content for this section. This appendix collects the bare-minimum prerequisites that the body of the book either assumes or alludes to without developing in full. The level of detail is calibrated as Chapter 1 lands. Topics:

- Sigma-algebras and measurability (sketched at the level needed)
- Limits, continuity, ε-δ
- Riemann vs Lebesgue integration (one paragraph)
- Real analysis essentials (Weierstrass, Bolzano–Weierstrass)
- Orthogonal matrices and the orthogonal group $O(n)$
- A note on conventions (we use X^T for transpose, etc.)

Placeholder content for this section. Each of these gets one short section. Pointers to the literature (Tao's analysis books, Folland) are given at the end of each section.

Annotated Reading List

Placeholder content for this section. This is not the bibliography (the actual bibliography is at the end of the book in BibTeX). This is the annotated *reading list: 1–3 sentences per source explaining what it does well and what to read it for. Pattern borrowed from Larry Wasserman's 'Bibliographic Remarks' sections in* All of Statistics.

Probability foundations

Placeholder content for this section. Casella & Berger, Feller, Billingsley.

Linear algebra

Placeholder content for this section. Strang, Horn & Johnson.

Random matrix theory

Placeholder content for this section. Tao's Topics in Random Matrix Theory *(free PDF), Anderson–Guionnet–Zeitouni, Bai & Silverstein, Couillet & Debbah.*

High-dimensional probability

Placeholder content for this section. Vershynin HDP, *Wainwright* HDS.

Spectral clustering

Placeholder content for this section. Von Luxburg's tutorial, Ng–Jordan–Weiss, Shi–Malik.

Tracy–Widom and the spectral edge

Placeholder content for this section. Original Tracy–Widom papers, Edelman & Sutton expository pieces, the Quanta Magazine article.

Applications

Placeholder content for this section. Connections to deep learning, finance, biology, short pointers.

Joseph Genzer's honors thesis

Placeholder content for this section. The structural seed of this book; covers the multiscale phase transition Mathemagics defers to Volume II.

Bibliography

[NRI14] NRICH Maths Project. *What is a Random Variable, Really?* https://nrich.maths.org/articles/what-random-variable-really. University of Cambridge. Accessed 2026. 2014.

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